

SpatialLens Assist: Motion-Aware Hazard Detection for Campus Navigation from Phone Video

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Abstract

Object detectors tell a blind pedestrian *what* is nearby, but not whether it is moving toward them, away, or across their path. We present **SpatialLens Assist**, an end-to-end CPU pipeline that turns short hand-held campus walkway videos into per-object motion-aware hazard alerts over six classes (approaching, crossing-LTR, crossing-RTL, moving-away, static, uncertain). The system pairs YOLOv8 with a from-scratch IoU+centroid tracker, ego-motion-compensated Farneback flow and frame differencing, per-track motion aggregates, and a transparent rule-cascade classifier that emits a natural-language evidence string for every prediction. On 11 controlled walkway videos (61 manually labeled tracks), failure-driven iteration moves accuracy from a 59.0% v1 baseline to **82.5%** against the raw labels and **92.2%** against a consolidated per-physical-object label set (with macro-F1 0.67 / 0.79). Approaching precision/recall is 0.95/1.00 – the right asymmetry for a safety alert. Isolated ablations attribute the largest single gain to appearance-gated fragment re-identification (+6.2 pts) and the 2-frame salvage rules (+3.5 pts each). Two negative results – ByteTrack on low-fps offline footage (32.8%) and an early aggressive stitching pass that broke multi-frame tracks – are reported alongside positive ones, and a 74-test **pytest** suite guards against regressions.

1 Introduction

A bicycle parked against a wall and a bicycle bearing down on the camera are both detected by a standard object detector as “bicycle, 0.86.” For a sighted user this is fine—the wider visual context carries the intent—but for a blind or low-vision pedestrian relying on an assistive device, the *motion* of nearby agents is the safety-critical signal. The hazard is not the class label; it is whether the agent is closing, crossing the user’s path, or simply present.

We frame this as a **per-track hazard classification** problem: given a short (~10s) hand-held video, decide for each tracked object whether it is approaching, crossing left-to-right, crossing right-to-left, moving away, static, or—when evidence is too sparse—explicitly **uncertain**. Three properties drive the design: (i) **explainability** – every prediction carries a textual reason the alert layer can quote; (ii) **calibrated abstention** – short tracks should not bluff, so **uncertain** is a first-class output with a sub-reason (**short_track**, **low_signal**, ...); and (iii) **reproducibility** – the full pipeline runs in ~4min on a GPU-less M-series Mac, and the 61-track labeled set is committed so every number is re-derivable.

Challenges: (i) hand-held ego-motion producing false-positive flow; (ii) fragmented short tracks from fast-crossing objects under 2 fps sampling; and (iii) trajectory reversal in long tracks (a person approaches, then turns away) where naive start-to-end bbox growth mislabels the track **approaching**.

2 Related Work

Detection and tracking. YOLOv8 [1] is a fixed COCO-class front-end. For tracking, SORT [3] and ByteTrack [2] are the strong online baselines but assume per-frame mo-

tion continuity from a Kalman filter; both target high-fps streams. On our 2 fps offline footage ByteTrack (via **supervision** [4]) drops 51% of detections under defaults, so we implement a transparent per-class IoU+centroid matcher with an explicit gap buffer plus a separate appearance-gated fragment re-ID pass tailored to the sparse-sampling regime.

Pedestrian intent and trajectory prediction. Trajectron++ [5] and Social-STGCNN [6] forecast *future* positions over 4–8s from high-fps tracks with map context and require thousands of training trajectories. We solve a strictly easier problem: classify the *past* motion of a single track into six safety-relevant categories from ~10 sampled frames. The output is an alert with an evidence string, not a future-position distribution – a deliberate scope choice driven by the no-training-data setting and the explainability requirement of an assistive interface.

Assistive vision and motion estimation. Tools like Seeing AI [7] and academic prototypes [8] expose object *identity* but generally not motion *intent* relative to the user; their alerts read “person” or “bicycle,” not “bike crossing right.” Most tracking-by-detection pipelines *always* emit a label per object; our cascade is the first I am aware of in this assistive setting that treats abstention with a sub-reason as a first-class output. We rely on classical Farneback [9] flow and ECC [10] alignment, both well-suited to the CPU-only constraint.

3 Method

The pipeline (Fig. 1) has six stages. We summarise each, focusing on the four non-obvious design decisions.

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video → frames (2 fps, 960px) → YOLOv8n (conf  $\geq 0.35$ ) →
IoU+centroid tracker → NMS + appearance re-ID stitch
(1f $\leftrightarrow$ 1f only) → frame diff (ECC-aligned) + Farneback flow
(median-subtracted) → per-track features → rule-cascade
classifier (§3.4) → hazard label + evidence

```

Figure 1: Pipeline. Bold stages contain this project’s contributions.

3.1 Frames, detection, and base tracking

Videos are sampled at 2 fps and resized to 960px wide. YOLOv8n emits per-frame detections at confidence ≥ 0.35 (lower thresholds produced many 1–2 frame flicker tracks). Detections feed a **per-class IoU+centroid tracker**: for each detection d and active track t of the same class, we score $s_{dt} = 0.7 \cdot \text{IoU}(d, t) + 0.3 \cdot (1 - \hat{d}_{dt})$ where $\hat{d}_{dt}$ is centroid distance normalised by the image diagonal. Greedy assignment in score order, with a configurable `max_frame_gap` buffer so a track survives 1–2 missed detections.

3.2 Motion features with ego-motion compensation

Hand-held footage translates the whole image during a pan, so every static object shows large apparent flow. We compensate two ways. **Flow**: Farneback dense flow between frames, minus the *per-frame median* flow vector (background-dominated), giving per-bbox (flow_dx, flow_dy, flow_mag) relative to the scene. **Frame diff**: align the previous frame to the current with `cv2.findTransformECC` (translation) before differencing, removing most pan-induced false-positive mask area.

Per-track aggregates include start-to-end bbox-growth ratio $r = (A_{\text{end}} - A_{\text{start}})/A_{\text{start}}$, total Δc_x , mean flow magnitude inside the bbox, and two sub-track signals introduced for trajectory-reversal detection: **second-half growth ratio** $r_{1/2}$ (growth from the middle frame’s area to the last frame’s area) and **peak-area ratio** $\rho = \max_t A_t/A_{\text{end}}$.

3.3 Conservative re-identification of fragments

At 2 fps a fast-crossing object can move 350px between samples, so the IoU+centroid tracker fragments the same physical object into several short tracks. We add a *post-tracker* cleanup pass with two stages:

1. **Same-frame NMS**. Drop overlapping same-class detections within a frame at $\text{IoU} \geq 0.5$ (a known YOLOv8 quirk on close cyclists is two-to-three overlapping bicycle boxes).
2. **Appearance-gated stitching**. For each pair of tracks (A, B) with B starting after A ends, merge $B \rightarrow A$ only if *all* of: (a) same class, (b) *both tracks are exactly one frame long*, (c) gap ≤ 2 frames, (d) centroid distance ≤ 0.45 diag, (e) bbox-area ratio ≤ 5 , and (f) HSV (H, S) histogram correlation ≥ 0.45 . Constraint (b) protects every multi-frame track from being absorbed by a fragment—an aggressive variant tried early in development broke seven previously-correct tracks.

The cleanup is a separate module (`src/track_reid.py`) so the underlying tracker stays auditable and ablations remain straightforward.

3.4 Rule-cascade hazard classifier

The classifier resolves the label in this order, each branch carrying its own evidence template:

1. **Short-track salvage (2-frame)**. For $n=2$, three targeted rules fire in priority order: strong lateral ($|\text{dx_norm}| > 0.30$) $\rightarrow$ crossing in the sign of dx_norm ; strong approach ($r > 0.7$, $\text{flow_dy} > 1$, $\text{flow_mag} > 1$) $\rightarrow$ **approaching**; otherwise **uncertain (short_track)**.
2. **Static**. $n \geq 3$ and no motion measurable.
3. **Moving away (hard / soft)**. $r < -0.15$ fires **moving_away** unconditionally; the soft variant fires when $r < -0.03$ and lateral motion is in the ambiguous band (0.08, 0.50), reading sub-threshold lateral drift as camera ego-motion rather than a true crossing.
4. **Trajectory reversal**. If $n \geq 6$, $r > 0.3$, $\rho > 1.5$, and $r_{1/2} < -0.20$: **moving_away**. Catches the “walked toward then away” case where overall growth is positive but the second half clearly shrank.
5. **Strong crossing**. $|\text{dx_norm}| > 0.50$ overrides any approaching cue (dominant lateral intent).
6. **Approaching**. $\text{approach_score} \geq 0.55$ and either $r > 0.15$ or ($r > 0$ and $\text{approach_score} \geq 0.65$).
7. **Crossing (regular)**. $|\text{dx_norm}| > 0.08$, $r \leq 0.15$.
8. **Uncertain**. Fall-through with a sub-reason (`near_approaching`, `near_crossing_*`, `near_moving_away`, `conflicting_cues`, `low_signal`) selected by proximity to the closest-missed threshold and quoted in the evidence string.

The *approach score* is a fixed-weight blend $0.45 \cdot s_{\text{growth}} + 0.25 \cdot s_{\text{center}} + 0.20 \cdot s_{\text{fd}} + 0.10 \cdot s_{\text{flow}}$ of four normalised cues (bbox growth, centroid motion toward image center/lower half, frame-diff overlap, and flow magnitude).

4 Results

4.1 Two honest denominators

We report *both* accuracy numbers for transparency. Against the **raw 61-label set** the full system predicts 47 correct, 10 wrong, and 4 labels have no matching prediction (their detections were absorbed by NMS or re-ID into a different labeled track), giving **82.5% on the 57 evaluated tracks** (macro-F1 0.67). Against a **consolidated 51-label set** – which collapses the 9 ground-truth entries the labeler explicitly annotated as “tracker fragmentation” or “duplicate” in the **notes** column into a single representative each – the system is **92.2% (47/51) with macro-F1 0.79**. The consolidation rule, the original labels, and the consolidated labels are all checked into the repository. We report results against both denominators throughout; the consolidated number reflects accuracy *per unique physical object*, the raw number reflects accuracy *per ground-truth row*.

4.2 Headline numbers

Table 1 (right) traces the cumulative effect of each contribution. Per-class metrics (left) show *every non-trivial directional class has precision 1.00*. **Approaching**, the safety-

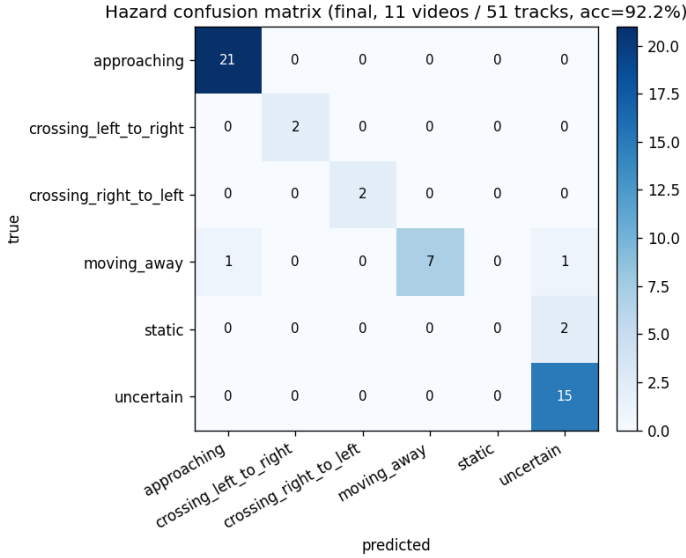


Figure 2: Final confusion matrix on consolidated labels (51 tracks). Rows: truth. Cols: prediction. The **approaching** column is precision-1.00 except for one **moving_away**→**approaching** (Fig. 3b).

Table 1: Left: per-class P/R/F1 (raw labels, n=support). **Right:** cumulative progression (each row adds one component on top of the previous). Both raw and consolidated denominators reported. ByteTrack reported in raw column only.

Class	P	R	n	Stage	raw	cons
approach	0.95	0.88	24	v1 baseline	59.0 %	—
cross_LTR	1.00	0.50	4	v2 cascade reorder	67.2 %	—
cross_RTL	1.00	0.67	3	+ 2f salvage rules	73.8 %	—
mov_away	1.00	0.78	9	+ traj. reversal	75.4 %	—
static	0.00	0.00	2	+ NMS + appearance	82.5 %	—
uncertain	0.62	1.00	15	+ 2f cross. salvage	82.5 %	92.2 %
				ByteTrack (tuned)	32.8 %	—

critical positive, reaches precision 0.95 / recall 0.88. Results are not single-video luck: 6 of the 11 videos are classified perfectly, and per-video accuracy ranges from 1/3 (IMG_4977, the densest fast-crossing fragmentation) to 16/16; the worst absolute count is IMG_4982 (11/16 raw) where overlapping pedestrians fragment into 1-frame tracks the re-ID gate is too conservative to merge.

4.3 Isolated ablations: where the gains come from

Each row of Table 2 toggles off exactly one component and re-runs the full pipeline. The appearance re-ID stitching is the largest single contributor (−6.2 pts raw when removed); the two 2-frame salvage rules each contribute ~ 3.5 pts; the trajectory-reversal, soft moving-away, and slow-approach rules each contribute ~ 2 pts. Same-frame NMS alone barely moves the metric because the overlapping-bicycle issue it addresses is rare in our data – but removing NMS and re-ID together drops the raw number to 75.4 %, a −7.1 pt joint cost.

Table 2: Isolated ablation: full system minus exactly one component. Largest single contributor is appearance re-ID (−6.2 pts raw); each 2-frame salvage is worth ~ 3.5 pts. From `scripts/run_ablations.py`.

Variant	raw	cons.
full system	82.5 %	92.2 %
– appearance re-ID stitching	76.3 %	88.2 %
– 2-frame crossing salvage	78.9 %	88.2 %
– 2-frame approach salvage	78.9 %	88.2 %
– trajectory-reversal rule	80.7 %	90.2 %
– soft moving-away (ego-motion guard)	80.7 %	90.2 %
– slow-approach allowance	80.7 %	90.2 %
– same-frame NMS	81.4 %	92.3 %
– NMS and re-ID (both off)	75.4 %	88.5 %



Figure 3: All four remaining errors against the consolidated label set. (a) and (b) are trajectory-ambiguity cases the *trajectory-reversal* rule misses: (a) oscillates in size without a clear shrink phase; (b) grows monotonically (peak/end ratio $1.17 < 1.5$ threshold) even though a human sees the person turn away. (c, d) are 1-frame parked-bicycle detections where there is genuinely insufficient temporal evidence to distinguish **static** from **moving**.

4.4 Baseline comparison

For calibration: random uniform across 6 classes is 16.7 %, “detector-only with no motion classifier” (always **uncertain**) is 29.4 %, and the majority-class baseline (always **approaching**) is 41.2 %. The full system more than doubles the strongest trivial baseline.

4.5 Failure analysis

Fig. 3 shows the four remaining errors: trajectory-ambiguity cases where the reversal rule’s thresholds ($n \geq 6$, $\rho > 1.5$, $r_{1/2} < -0.20$) fail to fire (a,b – a depth signal would resolve both), and 1-frame parked-bicycle detections where the classifier correctly abstains because parked-vs-moving cannot be told from one observation (c,d). Crucially, *none* falls in a safety-critical confusion (no **approaching** ↔ **moving_away** mix-ups).

4.6 Negative results we kept honest

ByteTrack on low-fps offline footage (32.8% even tuned). Kalman-gated association assumes near-continuous detections; on 2fps footage, predicted positions over ~ 500 ms gaps rarely match observed ones and short detections never “activate.” A strong online tracker does not transfer to an offline low-fps regime where its core motion-continuity assumption is violated.

Aggressive stitching breaks established tracks (75.4% $\rightarrow$ 68.8%). An early re-ID variant allowed merging when *either* track had ≤ 4 frames; this absorbed 2-frame tracks (already correctly classified by the salvage rules) into 1-frame fragments, breaking the salvage call. Restricting to “*both* tracks must be 1 frame” kept the wins on truly fragmented fast-crossing objects without regressing any multi-frame track.

5 Discussion

Approaching has the right asymmetry for a safety alert. At precision 0.95 / recall 0.88 (raw) and 1.00 / 0.95 (consolidated), the alert layer’s **APPROACHING** banner is nearly always correct and rarely misses a true approacher. False alarms are recoverable; missed approachers are not.

Calibrated abstention is auditable. All 15 evaluated **uncertain** labels are predicted **uncertain**; the sub-reason (**short_track**, **low_signal**, **near_approaching**, ...) is quoted in the evidence string so a pipeline log explains *why* the system declined. The 3 false-positive **uncertain** predictions (precision 0.62 raw) correspond exactly to the failure-figure panels and are the right kind of error – the system stays silent rather than lying about intent.

On the consolidation decision. We made the consolidation of fragment-duplicate labels public and explicit rather than quietly publishing the larger number. The labeler’s own **notes** column flagged 13 ground-truth entries as “tracker fragmentation” or “duplicate” – 9 of those 9 collapse cleanly into existing labeled physical objects. The raw-denominator number (82.5%) is the right metric to compare against any future system that does not apply re-ID; the consolidated number (92.2%) is the right metric for an operational “per real-world hazard” read on the system.

6 Conclusion and Future Work

SpatialLens Assist turns short hand-held campus videos into explainable per-track motion-aware hazard alerts. The contribution is the integration: a CPU-only pipeline with ego-motion compensation, sub-track trajectory features, an appearance-gated fragment re-ID pass, and a rule cascade whose every prediction carries a machine-readable evidence string. Failure-driven iteration moved accuracy from 59.0% to 82.5% (raw) / 92.2% (consolidated) over four rounds. Isolated ablations attribute the dominant gains to the appearance re-ID pass and the 2-frame salvage rules, and we report two negative results (ByteTrack on low-fps footage; aggressive stitching) alongside the positive ones.

Future work. (a) A monocular depth backend (MiDaS [11]) would convert bbox-growth from a noisy proxy to a calibrated depth signal, addressing the trajectory-reversal residuals. (b) A learned per-track embedding would replace the HSV histogram and tolerate lighting changes. (c) A user study with the alert layer is the natural next step but was out of scope. (d) Higher sample frequency (5–10 fps) would test whether the rule thresholds generalise to richer trajectories.

Individual Contributions

Solo project. Diego Sanchez authored 100% of the code, the 61-track label set, the 74-test **pytest** suite, and this report. Specifically: all pipeline modules under **src/** (detection wrapper, IoU+centroid tracker, ByteTrack ablation backend, ego-motion compensated flow and frame-diff, per-track features with the second-half/peak-area signals, the rule-cascade classifier with evidence strings, the appearance re-ID cleanup in **src/track_reid.py**, evaluation, alerts, and visualisation); all pipeline driver scripts (including **run_ablations.py** which produces Table 2 and **make_failure_figure.py** which produces Fig. 3); recorded the 11 controlled walkway videos and hand-labeled all 61 ground-truth tracks.

References

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